

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT: Artificial intelligence

PRACTICAL NO. 3

Aim: Decision Tree

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load the Iris dataset
df = pd.read_csv("iris.csv")

# Display the first five records
print("First five records:")
print(df.head())

# Select input features and target variable
X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
y = df['Species']

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("\nTraining records:", len(X_train))
print("Testing records:", len(X_test))

# Create the Decision Tree model using Gini Index
model = DecisionTreeClassifier(
    criterion='gini',
    random_state=42
)
```

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```
# Train the Decision Tree model
model.fit(X_train, y_train)

# Predict the classes for test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nDecision Tree Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

# Display confusion matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Display classification report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Visualize the confusion matrix
plt.figure(figsize=(6, 5))

classes = model.classes_

plt.imshow(cm)

plt.xticks(
    range(len(classes)),
    classes,
    rotation=30
)

plt.yticks(
    range(len(classes)),
    classes
)
```

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```
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.title("Decision Tree Confusion Matrix")
```

```
# Display values inside confusion matrix
for i in range(len(classes)):
    for j in range(len(classes)):
        plt.text(
            j,
            i,
            cm[i, j],
            ha='center',
            va='center'
        )
```

```
plt.tight_layout()
plt.show()
```

```
# Visualize the generated Decision Tree
plt.figure(figsize=(16, 9))
```

```
plot_tree(
    model,
    feature_names=X.columns,
    class_names=model.classes_,
    filled=True,
    rounded=True
)
```

```
plt.title("Generated Decision Tree")
plt.show()
```

```
# Display Feature Importance
importance = pd.Series(
    model.feature_importances_,
    index=X.columns
).sort_values(ascending=False)
```

```
print("\nFeature Importance:")
```

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```
print(importance)

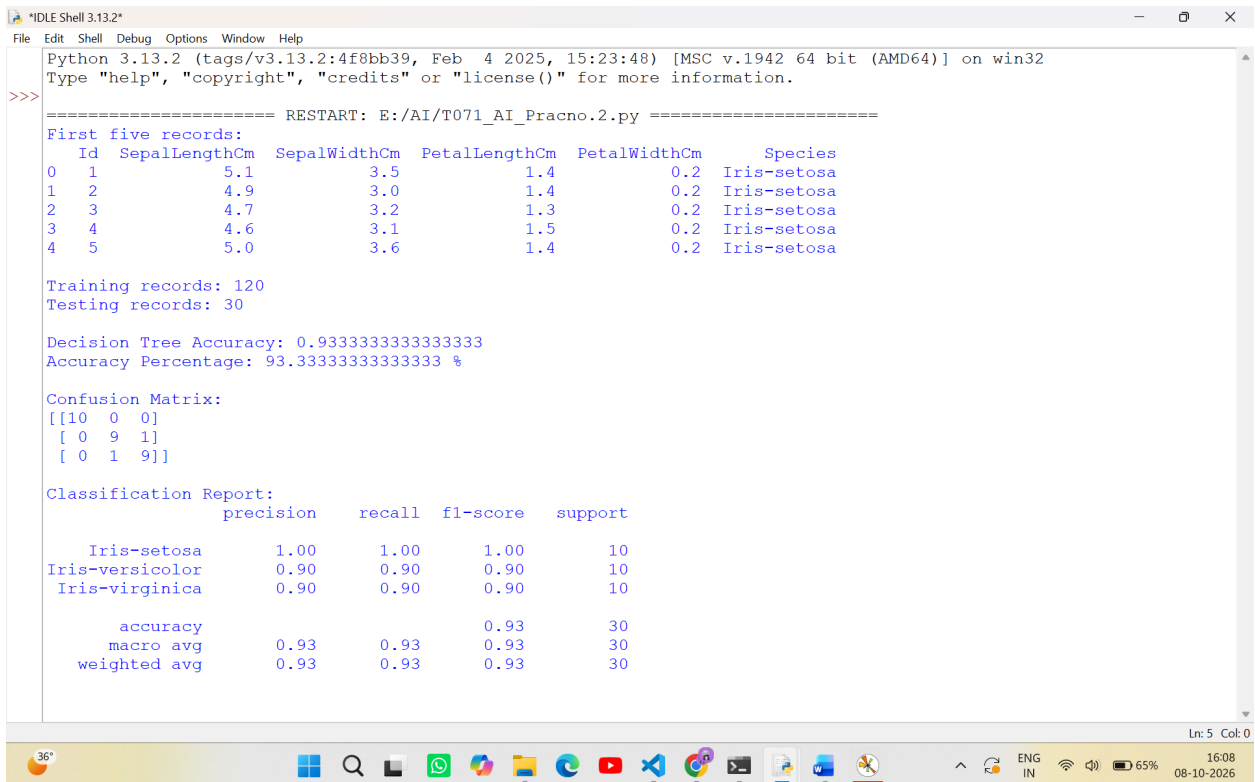
# Visualize Feature Importance
plt.figure(figsize=(8, 5))

plt.bar(
    importance.index,
    importance.values
)

plt.xlabel("Features")
plt.ylabel("Importance")
plt.title("Feature Importance in Decision Tree")
plt.xticks(rotation=20)

plt.tight_layout()
plt.show()
```

OUTPUT:



```
*IDLE Shell 3.13.2*
File Edit Shell Debug Options Window Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: E:/AI/T071_AI_Pracno.2.py =====
First five records:
  Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0  1             5.1             3.5             1.4             0.2  Iris-setosa
1  2             4.9             3.0             1.4             0.2  Iris-setosa
2  3             4.7             3.2             1.3             0.2  Iris-setosa
3  4             4.6             3.1             1.5             0.2  Iris-setosa
4  5             5.0             3.6             1.4             0.2  Iris-setosa

Training records: 120
Testing records: 30

Decision Tree Accuracy: 0.9333333333333333
Accuracy Percentage: 93.33333333333333 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  1  9]]

Classification Report:
              precision    recall  f1-score   support

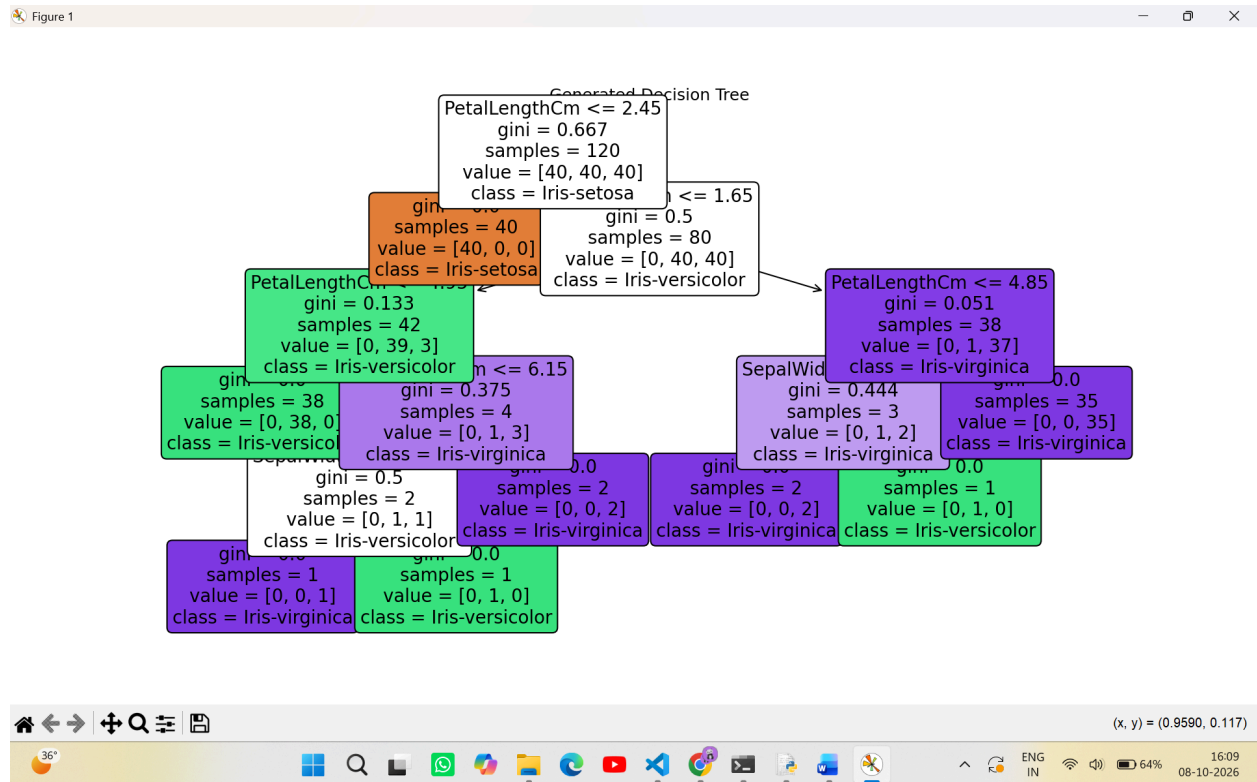
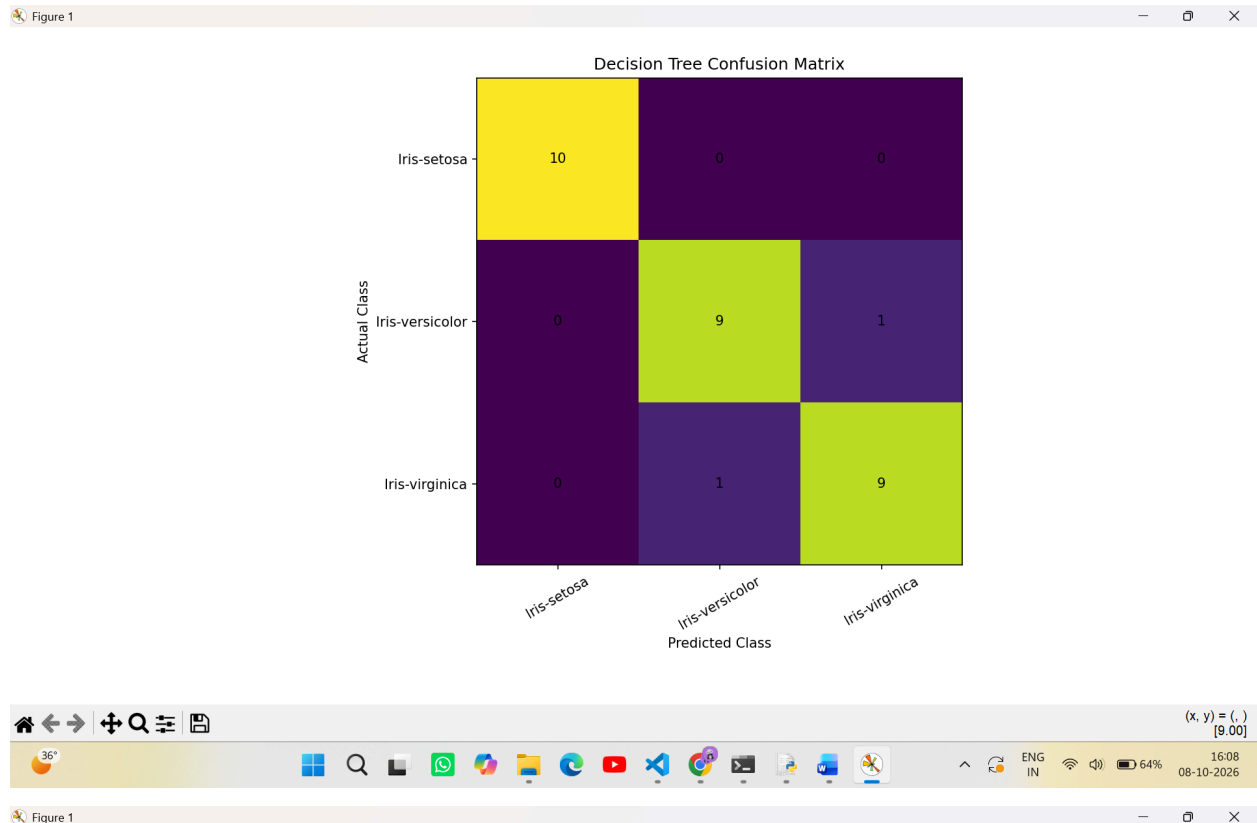
 Iris-setosa         1.00      1.00      1.00        10
 Iris-versicolor     0.90      0.90      0.90        10
 Iris-virginica      0.90      0.90      0.90        10

 accuracy          0.93
 macro avg         0.93
 weighted avg      0.93
```

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